

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 11-02-2026**

Time: 09.00Am to 12.00Pm **GTA-16** **Max. Marks: 186**

11-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P1)_GTA-16_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2019-P1-Model

Time: 3:00Hr's | IMPORTANT INSTRUCTIONS | Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 5 – 12)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 22)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 23 – 30)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 40)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 41 – 48)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

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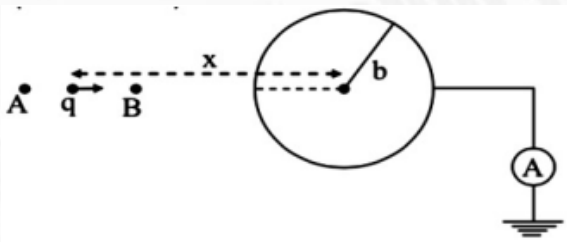
PHYSICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

1. If a charge 'q' is moving towards the centre of an earthed conducting sphere of radius 'b' with uniform velocity 'v'. Distance of two points A and B from centre of sphere are 3a and 2a. Conducting sphere is earthed with an ammeter and resistance in series (as shown)



Reading a ammeter at this instant

A) $q\frac{xV}{b^2}$ **B)** $q\frac{V}{b}$ **C)** $\frac{qbV}{x^2}$ **D)** Zero

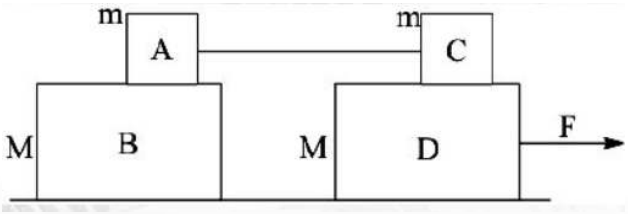
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2. A solid conducting sphere having a charge Q is surrounded by an uncharged concentric conducting hollow spherical shell. The potential difference between the surface of solid sphere and the outer surface of the hollow shell is V. If the shell is now given a charge of - 3Q, the new potential difference between the same two surfaces is

A) V **B)** 2 V **C)** 4 V **D)** $-2V$

3. Four blocks are arranged on a smooth horizontal surface as shown. The masses of the blocks are given (see the figure). The coefficient of static friction between the top and the bottom blocks is μ_s. What is the maximum value of the horizontal force F, applied to one of the bottom blocks as shown, that makes all the four blocks move with the same acceleration?

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A) $F_{max} = 2\mu_s mg\left(\frac{2m+M}{m+M}\right)$ **B)** $F_{max} = \mu_s mg\left(\frac{m+M}{2m+M}\right)$

C) $F_{max} = 2\mu_s mg\left(\frac{m+M}{2m+M}\right)$ **D)** $F_{max} = \mu_s mg\left(\frac{2m+M}{m+M}\right)$

4. ${}^{7}_{3}\text{Li} + {}^{1}_{1}\text{H} \rightarrow {}^{7}_{4}\text{Be} + {}^{1}_{0}\text{n}$ is endothermic. What is the minimum kinetic energy (in keV) of incident proton so that the reaction occurs? The Q value of the reaction is -1645 keV

A) 1880 **B)** 2125 **C)** 3350 **D)** 3840

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

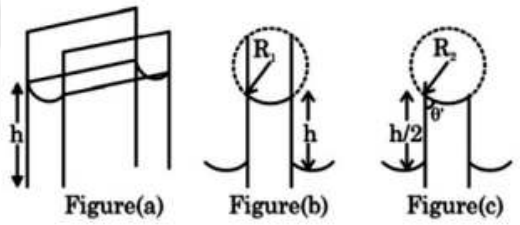
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Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

5. Two parallel plates at a distance d are submerged in water. (Surface tension = s and contact angle θ) Water has density ρ. The figure (a) shows the water rise and cross section view is in figure (b). Now the plates are pushed down so that the height of plates above water is h/s as in figure (c)



A) $h = \frac{2s\cos\theta}{\rho dg}$ **B)** $\cos\theta' = \frac{\cos\theta}{2}$ **C)** $R_1 = \frac{d}{2\cos\theta}$ **D)** $R_2 = \frac{d}{4\cos\theta}$

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6. A particle moves on a circular path of radius R, so that its angular velocity about a fixed point on the circumference of the circular path is constant and is equal to ω. Then

A) the acceleration of the particle is variable.

B) the acceleration of the particle is constant in magnitude and is equal to $4R\omega^2$

C) the acceleration of the particle varies in magnitude only

D) acceleration of the particle will be zero

7. The potential energy 'U' in Joule of a particle of mass1 kg , moving in $X-Y$ plane, obeys the law U = 3x + 4y, where (x, y)are the co-ordinates of the particle in meter. If the particle is at rest at (6, 4)at $t = 0$, then (Assuming there are no non-conservative forces acting on the particle)

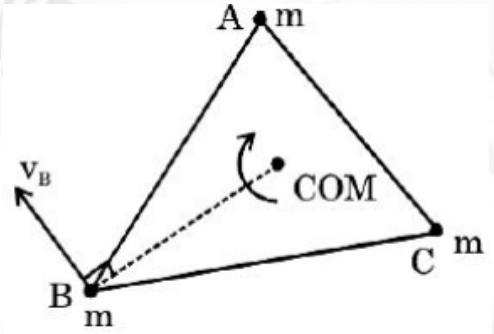
A) the particle has constant acceleration

B) the work done by the external forces, from the position of rest of the particle to the instant of crossing the X-axis is 50 J

C) the speed of the particle when it crosses the Y-axis is 10 ms^{-1}

D) the co-ordinates of the particle at time $t = 4s$ are (–18, –28)

8. Three small balls A, B and C of same mass 'm' are fixed by massless rods at the vertices of equilateral triangle with side L. The system is placed on a smooth horizontal surface and set 'm' rotation about the stationary COM with angular speed ω. At a certain instant ball B breaks off the system with a velocity $\vec{v}_B$ as shown in diagram where $|\vec{v}_B| = \frac{2\omega L}{\sqrt{3}}$



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A) The COM of A and C moves with a velocity $\frac{-\vec{v}_B}{2}$

B) The tension in rod connecting A and C after B breaks off is $m\omega^2\frac{l}{2}$

C) The speed of A just after B breaks off is $\frac{V_B}{2}$

D) The speed of A, after a time $\frac{\pi}{\omega}$ after B breaks off is $\frac{\sqrt{19}}{6}\omega l$

9. A syringe is filled with water. Its volume is 20 cm^3 and cross–section of its interior part is 4 cm^2. The syringe is held vertically such that its nozzle is at its top and its 100g piston is pressed by eternal agent and moves it with a constant speed. The ejected water has a initial upward velocity of 2 m/s and cross–section of the beam of water at the nozzle is 1 mm^2

A) The speed with which the piston is moving is 5 mm/sec.

B) The external force acting on the piston is 42 N.

C) The total work done by external agent in emptying the syringe is 0.095 J.

D) The external force remains constant during the process.

10. H like atom of atomic number Z is in an excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If it makes a transition to quantum state n, a photon of energy 40.8 eV is emitted. Identify the correct statement/s.

A) n = 2, Z = 16

B) n = 2, Z = 4

C) Ground state energy for this atom is –3481.6 eV

D) Ground state energy for this atom is –217.6 eV

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11. A hypothetical pulse is travelling along positive '*x*' direction on a taught string. The speed of pulse is 10 cm/s and shape of pulse at $t = 0$ is given as

$y = \frac{x}{6} + 1$ for $-6 \le x \le 0$

$= -x + 1$ for $0 \le x < 1 = 0$

For all other value of 'x'

If *x* and *y* are in cm, then

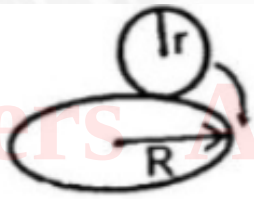
A) Vertical displacement of the particle at $x = 1$ cm at $t = 0.2$ s is $\frac{5}{6}$cm

B) Vertical displacement of the particle at $x = 1$ cm at $t = 0.2$ s is $\frac{4}{3}$cm

C) Transverse velocity of the particle at $x = 1$ cm at $t = 0.2$ s is $-\frac{5}{3}$cm / s

D) Transverse velocity of the particle at $x = 1$ cm at $t = 0.2$ s is $\frac{4}{3}$cm / s

12. A ring of radius '*r*' is moving without slipping on a circular track of radius *R*. Angular speed of the ring about its axis is ω. Ignoring the fact that the ring axis is inclined.



A) the angular acceleration of the ring is $\frac{\omega^2 r}{R}$

B) the angular acceleration of the ring is $\frac{\omega^2 (R - r)}{R}$

C) the angular velocity of centre of the ring about vertical axis passing through the centre of track is $\frac{r\omega}{R}$

D) the angular velocity of centre of the ring about vertical axis passing through the centre of track is $\frac{(R - r)\omega}{R}$

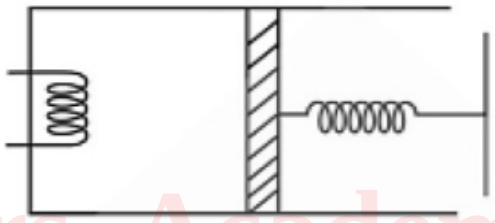
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SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

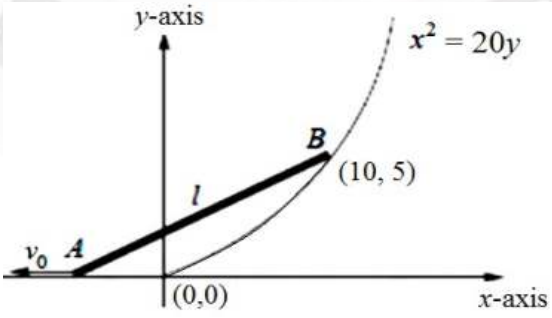
- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

13. An ideal monoatomic gas is confined by a spring loaded massless piston of cross section 8×10^{-3} m^2. Initially the gas is at 300 K and occupies a volume of 2.4×10^{-3} m^3 and the spring is in its relaxed state. The gas is heated by an electric heater until piston moves out without friction by 0.1 m . Force constant of the spring is 8000 N / m, atmospheric pressure 1×10^5 N / m^2, the cylinder and piston are thermally insulated, the piston and spring are massless and there is no friction between the piston and cylinder. Neglect the heat lost through the wire of the heater. The heat capacity of the heater is neglected.



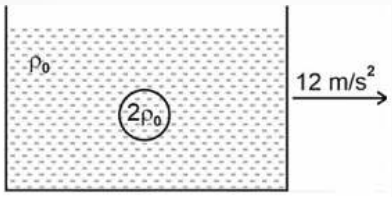
Find the ratio of heat supplied by the heater to the work done by the gas

14. A uniform rod has mass $m = 2$ kg and length $l = 13$m . One end of the rod is pulled with a constant velocity of $v_0 = 34$ m / s along a frictionless horizontal floor in the negative x direction. The other end is moving along a parabolic fixed curve. The equation of the parabola is $x^2 = 20y$. Find the angular velocity of the rod (in rad/s) when the end point 'B' is at (10, 5)

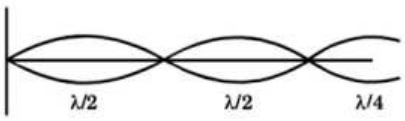


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15. A vessel containing liquid of density ρ_0 is accelerated horizontally with constant acceleration 12 m/s^2 in gravity free space. A small sphere of density $2\rho_0$ is released from rest with respect to the vessel. Magnitude of initial horizontal acceleration of sphere with respect to vessel is …… m/s^2

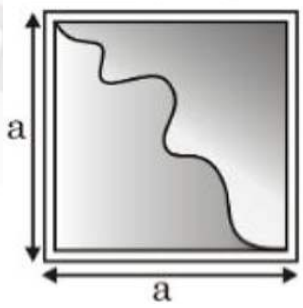


16. A string of length l fixed at one end and free at another end is vibrating in 5th harmonic. Maximum displacement of particle is A. Amplitude of oscillation of a particle at distance $\frac{l}{10}$ from free end is $\frac{A}{\sqrt{\alpha}}$ find value of α



17. The pitch of screwguage is 1mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line, zero of the circular scale is lying above the reference line. When the glass plate is placed between the studs the main scale reads three divisions and the circular scale reads 26 divisions. Calculate thickness of plate (in mm).

18. A thin soapy film is stretched at the square frame with a length a = 5.0 cm. A slender flexible thread with negligible mass is fastened at two diagonal points of the frame, separating soapy film in two regions as shown at the picture below. The film on one side is pierced and the thread takes a circular shape. The length of the thread is π a/2 and it is inextensible. Find the tension (in mN) in the thread if the surface tension of the soap film is 3mN/metre.



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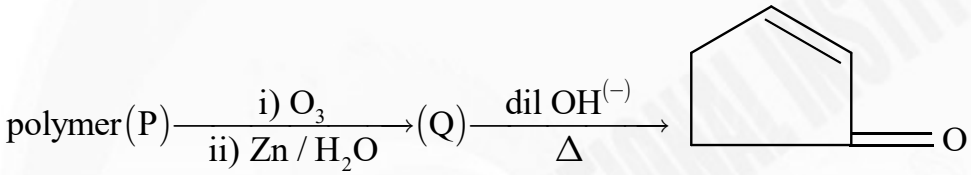
CHEMISTRY **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
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19.



Incorrect for above reaction sequence is

A) (P) exist in both cis and trans forms

B) (P) is a 1, 4-addition polymer of 2-methylbuta-1, 3-diene

C) (Q) can show both tollen's and Iodoform test

D) Q gives positive Iodoform test but no reaction with $NaHSO_3$

20. Which of the following statements is correct ?

A) Potassium ferricyanide obeys EAN rule and it give white ppt. with $AgNO_3$

B) Cyanocobalamine is used to treat anti pernicious anaemia factor and it give blue bead in borax bead test.

C) VBT can distinguish between weak and strong ligands accurately.

D) $[Co(EDTA)]^-$ contain five 5-membered ring and it is optically inactive

21. Which of the following is true?

A) Amongst noble gases, addition of electron is the easiest for Ne.

B) First electron gain enthalpy is endothermic for element having configuration $[Ne]3s^2 3p^3$

C) Amongst Li^-, Be^-, B^-, C^- the least stable is Be^-.

D) First electron gain enthalpy is less exothermic for Te than O

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22. In diamond, carbon atoms occupy FCC lattice points as well as alternate tetrahedral voids. If edge length of the unit cell is 356 pm, then radius of carbon atom is

A) 77.07 pm **B)** 154.14 pm **C)** 251.7 pm **D)** 89 pm

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

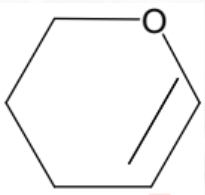
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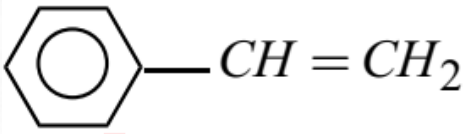
Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
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The Marking scheme to evaluate Answer to each question will be :
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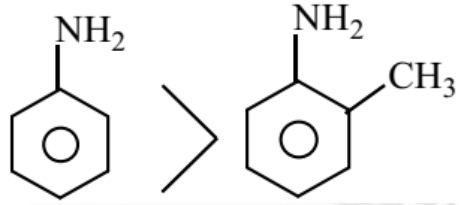
23. Which of the following is (are) more reactive then 2-Butene for electrophilic addition reaction

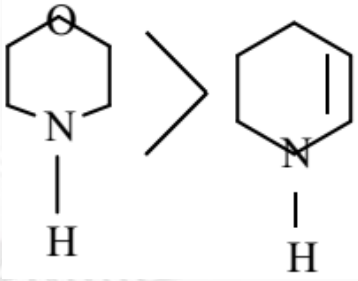
A) 

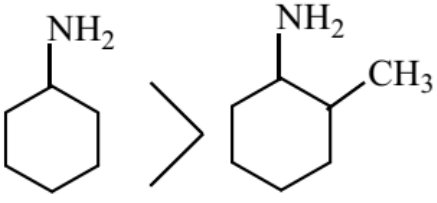
B) 

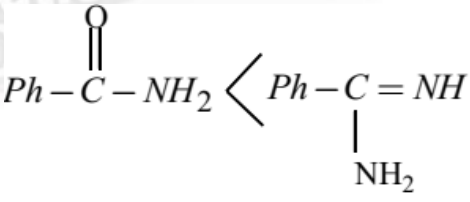
C) $CH_2 = CH - CH_2 - CH_3$ **D)** $CH_3 - C \equiv C - CH_3$

24. Correct order of basic strength of the following are :

A) 

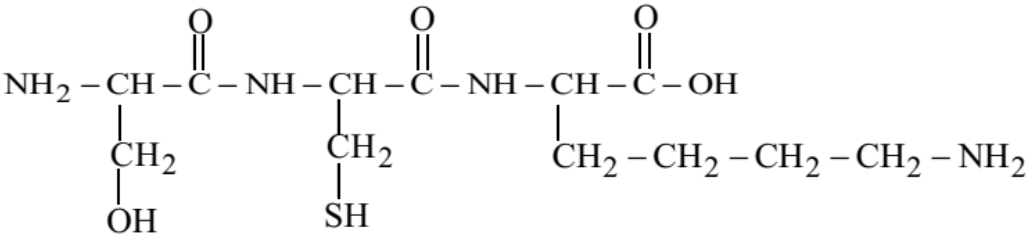
B) 

C) 

D) 

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25. Which of the following statement(s) is/are correct



Pentide

A) Above is a tripeptide with 2-peptide linkages

B) Peptide when subjected to Lassaigne's test with $FeCl_3$, gave blood red colour

C) One of the amino acid formed on hydrolysis is basic amino acid

D) One of the amino acid formed on hydrolysis is acidic amino acid

26. Which of the following statement(s) is/are incorrect?

A) The element with $[Xe]4f^{14}5d^1\ 6s^2$ belongs to 6th period and Group 3 of the modern periodic table

B) Electronic configuration of thorium (Z = 90) is $[Rn]5f^1 6d^1\ 7s^2$

C) Among $[V(CO)_6]^-$, $[Mn(CO)_6]^+$, $[Fe(CO)_5]$ and $[Ni(CO)_4]$; 'carbon – oxygen' bond length is maximum in $[Mn(CO)_6]^+$

D) Both permanganate and manganate ions are diamagnetic

27. Which of the following statement(s) is/are correct?

A) Cu^{2+} cation is reduced from its higher oxidation state +2 to +1 by both excess KI solution and excess of KCN solution

B) $Fe_2(SO_4)_3$ on heating gives only Fe_2O_3 and SO_2

C) $FeSO_4$ on heating gives Fe_2O_3, SO_2 and SO_3

D) Both Ag^+ and Pb^{+2} salt solutions gives white precipitate separately by adding excess KCN solution

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28. Which among the following is/are emulsifiers for oil in water type emulsion

A) Proteins **B)** Lamp black

C) Heavy metal salts of fatty acids **D)** Gums

29. Which of the following has ΔS_{sys} positive

A) formation of non ideal solution showing negative deviation from the Raoult's law

B) formation of an ideal solution

C) Conversion of diamond to graphite

D) Defect formation in a perfect crystal

30. Among the following, the correct statement(s) for electrons in an atom is/are

A) Uncertainty principle rules out the existence of definite paths for electrons.

B) The energy of an electron in 2s orbital of an atom is lower than the energy of an electron that is infinitely far away from the nucleus

C) According to Bohr's model, the most negative energy value for an electron is given by n = 1, which corresponds to the most stable orbit.

D) According to Bohr's model, the magnitude of velocity of electrons increases with increase in value of 'n'

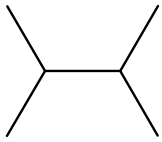
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SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

31. The number of $-CH_2-$ (Methylene) groups in the final product formed from the following reactions sequence is ……



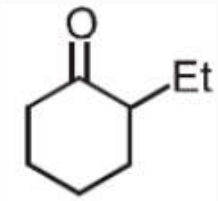
$$\xrightarrow[(3)(i)O_3\ \ (ii)Zn/H_2O]{\begin{matrix}(1)\ Br_2/\Delta(mono)\\(2)\ alc.\ KOH/\Delta\end{matrix}} (major\ product)$$

$$(4)\ conc.\ H_2SO_4/\Delta$$

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32. Total number of β -Keto monocarboxylic acids (including stereoisomers) which on heating give



33. 9 mole of equimolar mixture of Ferric oxalate and ferrous oxalate reacts with acidified $KMnO_4$ for complete oxidation. Find the number of mole of $KMnO_4$ required for it. (Write your answer to nearest whole no.)

34. How many of the following combination(s) is/are CORRECT?

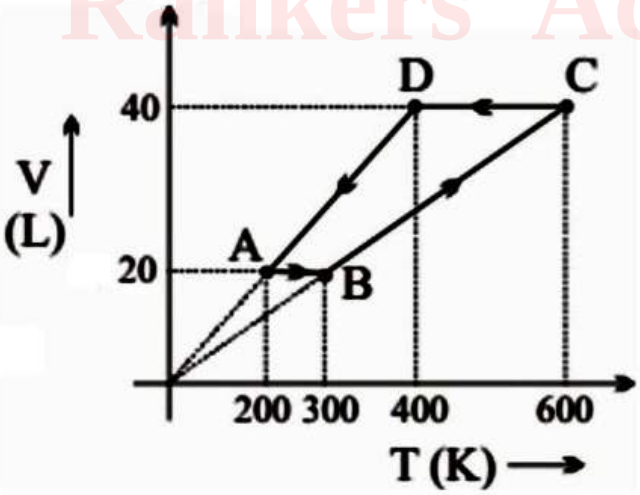
i) $PbCrO_4$ – Yellow ppt., ii) $Cu_2[Fe(CN)_6]$ – Chocolate brown colour ppt.,

iii) ZnS - Black ppt., iv) NiS - White ppt.,

v) $CaCl_2$ - Brick red flame on flame test, vi) $Mg(NH_4)PO_4$ - White ppt.

35. For a gas obeying P(V – b) = RT a graph is plotted between Volume (V lit.) on Y-axis and Temperature (K) on X-axis at a constant pressure of 0.04105 atm. Find the slope of the line?

36. Calculate the magnitude of work done (in atm L) for the given cyclic process involving 1 mol of an ideal gas. [Given : R = 0.08 atm L/mol/K]



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MATHEMATICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

37. Number of ordered pair (a, b) from the set A = {1, 2, 3, 4, 5} so that the function $f(x)=\frac{x^3}{3}+\frac{a}{2}x^2+bx+10$ is an injective mapping $\forall x \in R$:

A) 13 **B)** 14 **C)** 15 **D)** 16

38. Three *a's*, three *b's* and three *c's* are placed randomly in a 3×3 matrix. The probability that no row or column contain two identical letters can be expressed as $\frac{p}{q}$, where p and q are coprime then (p + q) equals to :

A) 151 **B)** 161 **C)** 141 **D)** 131

39. Number of points of discontinuity of $f(x)=\left\{\frac{x}{5}\right\}+\left[\frac{x}{2}\right]$ in $x\in[0,100]$ is/are (where [.] denotes greatest integer function and {.} denotes fractional part function)

A) 50 **B)** 51 **C)** 52 **D)** 61

40. If 'M' be a square matrix of order 3 such that |M| = 2, then $\left|\text{adj}\left(\frac{M}{2}\right)\right|$ equals to :

A) $\frac{1}{2}$ **B)** $\frac{1}{4}$ **C)** $\frac{1}{8}$ **D)** $\frac{1}{16}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**
Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

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41. Consider the line $L_1 : x = y = z$ and the line $L_2 : 2x + y + z - 1 = 0 = 3x + y + 2z - 2$, then:

A) The shortest distance between the two lines is $\frac{1}{\sqrt{2}}$

B) The shortest distance between the two lines is $\sqrt{2}$

C) Plane containing the line L_2 and parallel to line L_1 is $z - x + 1 = 0$

D) Perpendicular distance of origin from plane containing line L_2 and parallel to line L_1 is $\frac{1}{\sqrt{2}}$

42. For the equation $\frac{e^{-x}}{1+x} = \lambda$ which of the following statement(s) is/are correct?

A) When $\lambda \in (0, \infty)$ equation has 2 real and distinct roots

B) When $\lambda \in (-\infty, -e^2)$ equation has 2 real and distinct roots

C) When $\lambda \in (0, \infty)$ equation has 1 real root

D) When $\lambda \in (-e, 0)$ equation has no real root

43. Let z_1, z_2 and z_3 be three distinct complex numbers, satisfying $|z_1| = |z_2| = |z_3| = 1$. Which of the following is/are true?

A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z - z_1}{z - z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$

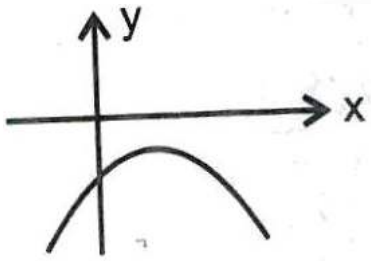
B) $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$

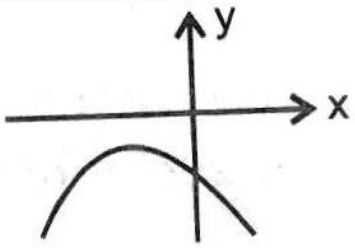
C) $\text{Im}\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 . z_2 . z_3}\right) = 0$

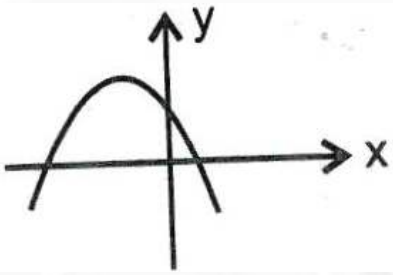
D) If $|z_1 - z_2| = \sqrt{2}\ |z_1 - z_3| = \sqrt{2}\,|z_2 - z_3|$, then $\text{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$

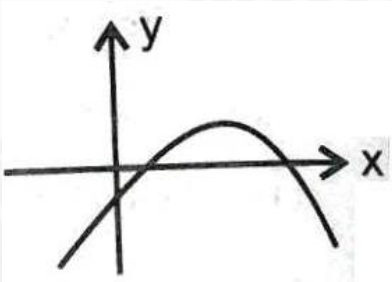
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44. If a, b, c $\in$ R, then for which of the following graphs of the quadratic polynomial $y = ax^2 - 2bx + c(a \neq 0)$; the product (abc) is negative?

A) 

B) 

C) 

D) 

45. The equation of the sides of the triangle having (3, -1) as a vertex and x – 4y + 10 = 0 and 6x + 10y – 59 = 0 as angle bisector and as median respectively drawn from different vertices, are :

A) 6x + 7y – 13 = 0 **B)** 2x + 9y – 65 = 0

C) 18x + 13y – 41 = 0 **D)** 6x – 7y – 25 = 0

46. The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} {}^{100}C_k (x-2)^{100-k} 3^k$ is also equal to :

A) number of ways in which 50 identical books can be distributed to 100 students, if each student can get atmost one book.

B) number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together.

C) number of dissimilar terms in $(x_1 + x_2 + x_3 + ... + x_{50})^{51}$

D) $\frac{2.6.10.14.....198}{50!}$

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47. The value of definite integral :

$$\int_{-2014}^{2014} \frac{dx}{1+\sin^{2015}(x)+\sqrt{1+\sin^{4030}(x)}}$$ equals :

A) 0 **B)** 2014 **C)** $(2014)^2$ **D)** 4028

48. Given $\frac{\underbrace{1000.....01}_{\text{n zeroes}}}{\underbrace{1000.....01}_{(n+1)\text{ zeroes}}} < \frac{\underbrace{1000.....01}_{\text{m zeroes}}}{\underbrace{1000.....01}_{(m+1)\text{ zeroes}}}$, then which of the following is true?

A) $m+1<n$ **B)** $m<n$ **C)** $m<n+1$ **D)** $m>n+1$

SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

49. Let f(x) be a differentiable function in $[-1,\infty)$ and $f(0)=1$ such that

$\lim_{t\to x+1} \frac{t^2 f(x+1)-(x+1)^2 f(t)}{f(t)-f(x+1)}=1$. Find the value of $\lim_{x\to 1} \frac{\ln(f(x))-\ln 2}{x-1}$

50. Let $f(x)=x^2+2x-t^2$ and $f(x)=0$ has two roots $\alpha(t)$ and $\beta(t)$ $(\alpha<\beta)$ where 't' is a real parameter. Let $I(t)=\int_{\alpha}^{\beta} f(x)\,dx$. If the maximum value of I(t) be λ and $|\lambda|=\frac{p}{q}$ where 'p' and 'q' are relatively prime positive integers. Find the product (pq)

51. The equation of transverse axis of hyperbola (passing through origin) having asymptotes $3x-4y-1=0$ and $4x-3y-6=0$ is $ax+by-c=0$, $a, b, c \in N$ and gcd(a, b, c) = 1, then the value of (a + b + c) = ……..

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52. The complete set of values of 'x' satisfying the inequality $\sin^{-1}(\sin 5) > x^2 - 4x$ is $\left(2-\sqrt{\lambda-2\pi}, 2+\sqrt{\lambda-2\pi}\right)$, then $\lambda =$

53. Consider the statics of two sets of observations as follows:

	Size	Mean	Variance
Observation – I	10	2	2
Observation – II	n	3	1

If the variance of combined set these two observations is $\frac{17}{9}$ then n =

54. For any finite set S, let |S| denote the number of elements in S. The number of ordered pairs (A, B) such that A and B are (not necessarily distinct) subsets of {1, 2, 3, 4, 5} that satisfy $|A|.|B| = |A \cap B|.|A \cup B|$ is λ, then number of positive integral divisors of λ

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